A/B Testing & Statistical Analysis

By Ayushi Singh

Data Analyst | Python, SQL, Power BI, Excel

An end-to-end experimentation project: fundamentals, terminology, errors, statistical methods, and a complete worked example - from hypothesis and sample-size calculation to a ship / no-ship decision.

Topics covered: 1. A/B Testing Fundamentals 2. Use Cases Across Industries 3. How an A/B Test Works 4. Experiment Flow & Terminology 5. Errors in A/B Testing 6. Statistical Methods for Analysis 7. Real-World A/B Test Example with Analysis 8. Bonus: Resume Phrasing & Interview STAR Method

Honesty note: This is a simulated project built to demonstrate an enterprise-scale experimentation pipeline. It is not real company traffic. In an interview, say so plainly - interviewers care far more about how cleanly you think about experimentation than whether the website had real users.

Structure note: The section order and the worked example (checkout redesign, 5% baseline, 1-point MDE) follow the teaching example in 'AB Testing and Statistical Analysis by Rishabh Mishra', shared via Google Drive. The wording, SQL/Python code, and calculations in this document are my own work, and the confidence interval below is calculated correctly from the data (see Section 7, Calculation Note).

1. What is A/B Testing?

A/B testing is a randomized method of comparing two or more versions to see which one performs better. It helps in making data-driven decisions instead of launching a change for everyone on a guess.

- Control group (Version A): the existing version. It is the baseline.
- Treatment group (Version B): the new version you want to test.

Types of A/B Tests

- Standard A/B Tests - Compare two versions (A and B), for one variable.
- A/B/n Tests - Compare multiple versions (A, B, C, etc.), for one variable.
- Multivariate Test (MVT) - Compare multiple variables simultaneously. MVT needs much larger samples, because every combination must be tested.

Quick Example

A company wants more people to buy products, so the analytics team has an idea: 'What if we change the background colour of the checkout page from blue to orange?' Instead of changing it for everyone immediately, we run an experiment. Randomly divide users into two groups: Group A sees the existing version (blue, control), and Group B sees the new version (orange, treatment). Measure the results: compare how many users in each group actually purchase. If Group B generates more purchases, the new colour might be better. However, we must check whether the difference is statistically significant before concluding that the new design truly helped.

2. A/B Test Use Cases Across Industries

- Product and Website - CTA button text (primary metric: conversion rate); simplified onboarding page designs (primary metric: signup completion rate).
- Marketing and Advertising - Email subject lines (primary metric: click-through rate); ad creatives or landing page designs (primary metric: conversion rate).
- E-commerce and Retail - Free-shipping thresholds (primary metric: profit per visitor); discount offers (primary metric: purchase conversion rate).
- Subscription and SaaS - Monthly vs annual pricing page layouts (primary metric: paid subscription conversion rate); 7-day free trial vs 14-day free trial (primary metric: free-to-paid conversion rate).
- Mobile Apps and Gaming - Different game tutorials (primary metric: tutorial completion rate); push notification messages (primary metric: 7-day retention rate).
- Search Engines and Recommendation Platforms - Search-ranking algorithms (primary metric: successful search rate); personalised vs generic recommendations (primary metric: click-through rate).

The lesson: every use case names one primary metric first. If you cannot name it, you are not ready to run the test.

3. How Does an A/B Test Work?

As a Data Analyst, you are responsible for defining metrics, validating the data, writing SQL queries, calculating results, interpreting statistical tests, and presenting recommendations to stakeholders.

- Step 1: Define the business problem. Example: increase the purchase conversion rate.
- Step 2: Create a hypothesis. Example: the new checkout page will increase purchases.
- Step 3: Randomly assign users. Group A gets the old page; Group B gets the new page. Set up the experiment so each user stays in the same group.
- Step 4: Collect and analyse data. Calculate conversion rates, run the statistical analysis, and compare results.
- Step 5: Make a decision. Launch the change, reject it, or investigate further, based on the evidence - not on the first favourable day.

4. Experiment Flow & Terminology

- Control group (A): the group that receives the existing version. It acts as the baseline for comparison.
- Treatment group / Variant (B): the group that receives the new experience you want to test.
- Hypothesis: a statement you want to test. Null hypothesis (H0): there is no difference in the true conversion rates of Group A and Group B. Alternative hypothesis (H1): there is a difference in their true conversion rates.
- Minimum Detectable Effect (MDE): the smallest difference your experiment is designed to reliably detect at a specified significance level and power. A small improvement is harder to detect than a large one, so it typically requires a larger sample size.
- Primary metric: the main outcome used to judge the experiment's success, such as purchase conversion rate.
- Guardrail metrics: metrics that should not get worse while improving the primary metric. For example: average order value, refund rate, page load time, or complaint rate.
- p-value: the probability of observing a difference at least as large as the one seen, if the null hypothesis were true. If p-value <= 0.05, the result is statistically significant; if p-value > 0.05, it is not.
- Statistical significance: evidence that the observed difference is unlikely to be explained by random variation alone, under the assumptions of the test.
- Significance level (alpha): a predetermined risk threshold for a false positive (Type I error), most commonly set at 0.05. (1 - alpha) is the confidence level, i.e. 95%.
- Statistical power: the probability that your experiment detects an effect of a specified size when that effect really exists. Power = 1 - beta. If you choose 80% power, beta = 0.20: an 80% chance of detecting the effect, and a 20% Type II error probability for that effect. Higher power reduces the risk of missing a meaningful improvement, at the cost of a larger sample size.
- Randomisation: assigning users to groups randomly to reduce selection bias and make the groups comparable.
- Sample Ratio Mismatch (SRM): when the observed allocation differs unexpectedly from the planned allocation - for example, you expected a 50/50 split but observed 60/40. Investigate tracking, assignment logic, eligibility rules, or data pipelines before trusting the results.
- Peeking: repeatedly checking results and stopping the experiment when a favourable p-value appears. This inflates false-positive rates if the analysis does not account for repeated testing.
- Standard error: measures the uncertainty in a statistic caused by sampling variation. We do not test every user in the world - we test a sample, a slice of the audience, so every estimate carries uncertainty.

5. Errors in Experimentation

Type I Error (False Positive)

You conclude that the new version works, even though there is no real improvement. Example: the new checkout is actually no better than the old one, but your experiment produces a result that looks statistically significant, so you launch the new page. Setting alpha to 5% (0.05) controls the long-run false-positive rate under the null hypothesis. In simple words: you launch a change that has no true effect.

Type II Error (False Negative)

A real improvement exists, but your experiment fails to detect it. Example: the new checkout genuinely improves conversion from 5% to 6%, but your sample is too small. The test returns a non-significant result, so you conclude there is insufficient evidence of an improvement and miss a valuable opportunity. This is a Type II error, represented by beta. In simple words: you miss a real improvement.

How Sample Size, MDE, Error & Power Connect

- 1. Define the experiment: baseline conversion rate and MDE.
- 2. Choose statistical settings: significance level alpha and power (1 - beta).

- 3. Calculate required sample size and days to run: how many users each group needs at minimum.
- 4. Run and analyse the experiment: estimate the effect and evaluate the statistical evidence.

6. Statistical Methods to Analyse Test Results

- Z-test: checks whether the proportions of two groups with large sample sizes differ significantly. It calculates a Z-score, how far a value is from the mean in standard deviations (critical value about 1.96 for 95% confidence). Z-tests are for percentages or proportions. Examples: conversion rate, click-through rate.
- T-test: checks whether the means of two groups differ significantly. Used for averages or continuous metrics. Examples: average order value, revenue per user, session duration.
- Chi-square test: checks whether categorical variables are associated, or their proportions differ from what is expected. Used when data is split into groups or categories. Examples: a multi-option preference survey (four app colour themes, preferences by demographic); a browser/device distribution check (is the user distribution equal and random across mobile, tablet, and desktop?).

7. Real-World Example: Purchase Conversion Rate Experiment

Business scenario: An e-commerce company wants to increase purchases by redesigning its checkout page.

Step 1. Define the Objective and Hypothesis

- Objective: increase purchase conversion rate.
- Control (A): existing checkout page.
- Variant (B): redesigned checkout page.
- Primary metric: purchase conversion rate = users who purchase / eligible users exposed to the checkout page.
- Guardrail metrics: average order value, payment failure rate, checkout errors, and complaint rate.

Null hypothesis (H0): the redesigned checkout page does not change the true purchase conversion rate compared with the existing page. H0: conversion rate B = conversion rate A.

Alternative hypothesis (H1): the redesigned checkout page changes the true purchase conversion rate compared with the existing page. H1: conversion rate B is not equal to conversion rate A. This is a two-sided test: we would act on a significant drop as well as a significant lift.

Step 2. Set Experiment Parameters

Assume historical data shows a 5% purchase conversion rate.

- Baseline conversion: 5%
- MDE: 1 percentage point (5% to 6%)
- Significance level (alpha): 5%
- Statistical power: 80%
- Allocation: 50% A / 50% B

Using these assumptions, a two-proportion sample-size calculation requires approximately 8,160 users per group (about 16,320 total) as the minimum target. You can calculate this with a third-party tool, or manually with the formula below.

```
n per group = [ Z(1-alpha/2) * sqrt(2 * p-bar * (1-p-bar))
              + Z(power) * sqrt(p1*(1-p1) + p2*(1-p2)) ]^2 / (p2-p1)^2
p1 = 0.05, p2 = 0.06, p-bar = 0.055, Z(0.975) = 1.96, Z(0.80) = 0.84
n = 8,158 per group (rounded in the teaching example to ~8,160)
```

Step 3. Run the Experiment

- Randomly assign eligible users 50/50 to A or B.
- Keep each user's assigned version consistent for the whole experiment.
- Run until the planned sample size and experiment duration are met.
- Validate tracking, randomisation, and sample-ratio balance (SRM check) before looking at results.
- Avoid stopping early just because results look favourable.

Step 4. Analyse the Results

Suppose the experiment completes with these results:

- Group A (Existing): 10,000 users, 500 purchases, conversion rate 5%
- Group B (Redesigned): 10,000 users, 600 purchases, conversion rate 6%

Calculate the Impact

- Absolute lift: 6% - 5% = 1 percentage point
- Relative lift: (6% - 5%) / 5% = 20%
- Statistical test: two-proportion z-test, z-score = 3.10
- p-value: approximately 0.002
- 95% confidence interval for the lift: +0.37 to +1.63 percentage points (see Calculation Note below). Since it is

entirely above zero, the data support a positive improvement.
Since p-value (0.002) is below alpha (0.05), and the entire confidence interval is above zero, the result is statistically significant.

Calculation Note - Why the Interval Here Differs from the Teaching Deck

The teaching deck states a 95% interval of +0.63 to +1.37 percentage points for this example. That interval does not follow from the standard Wald calculation on 500/10,000 vs 600/10,000. The standard error for the difference is sqrt(0.05 x 0.95 / 10,000 + 0.06 x 0.94 / 10,000) = 0.00322 (0.322 points). The 95% interval is therefore 1.00 point +/- 1.96 x 0.322 = +0.37 to +1.63 points. The conclusion is unchanged - the interval is entirely above zero - but the correct interval is wider, and a wider interval is the honest one to quote. Verified with Python/SciPy: chi-square = 9.620, p = 0.00192, z = 3.102.

Step 5. Interpret and Recommend

- Result: Group B performed better.
- Conversion improved by 20% relative to Group A.
- The observed improvement is statistically significant.
- Check guardrail metrics to ensure the redesign did not harm order value or payment success.
- If results and business impact remain favourable, recommend a staged rollout and continued monitoring - not an instant 100% switch with no monitoring.

How to Build a 95% Confidence Interval

A confidence interval takes your single lift number (1 point) and wraps a safe boundary around it to account for uncertainty. Statisticians take the experiment's centre point (1 point) and add/subtract the standard error multiplied by 1.96 (the multiplier for 95% confidence). If that whole interval sits above zero, the data support a positive improvement; if it crosses zero, you cannot rule out no effect, even if the point estimate is positive.

Analysis Code (Python and SQL)

Python - two-proportion z-test and confidence interval:

```
from scipy import stats
import math
# 500/10000 vs 600/10000
chi2, p, _, _ = stats.chi2_contingency([[500, 9500], [600, 9400]], correction=False)
# chi2 = 9.620, p = 0.00192
pc, pv, n = 0.05, 0.06, 10000
se = math.sqrt(pc*(1-pc)/n + pv*(1-pv)/n)  # 0.00322
diff = pv - pc  # 0.01
print(diff - 1.96*se, diff + 1.96*se)  # 0.0037 to 0.0163
```

SQL - conversion by variant, one row per visitor, assigned once:

```
SELECT variant,
       COUNT(DISTINCT visitor_id) AS users,
       SUM(CASE WHEN purchased = 1 THEN 1 ELSE 0 END) AS purchases,
       1.0 * SUM(CASE WHEN purchased = 1 THEN 1 ELSE 0 END)
         / COUNT(DISTINCT visitor_id) AS conversion_rate
FROM checkout_experiment
GROUP BY variant;
```

8. Bonus: Add A/B Testing as a Skill in Your Resume

Use these as phrasing templates for how to describe experimentation work. Do not copy them as claimed results unless you actually ran that test and can defend the numbers in an interview.

- Product and Website - Checkout Page Redesign: Analysed checkout funnel drop-offs and user behaviour, partnered with product and engineering teams to optimise the checkout experience, and evaluated results through A/B testing, increasing checkout conversion rate by 12% and reducing cart abandonment by 8%. [Template example from the teaching deck.]
- Marketing and Advertising - Landing Page Optimisation: Evaluated landing page performance across traffic sources and audience segments, conducted A/B tests on page layouts and CTAs, and identified high-performing designs that increased landing page conversion rate by 15% and reduced cost per acquisition (CPA) by 10%. [Template example.]
- E-commerce and Retail - Discount Offer Optimisation: Analysed customer purchase behaviour and discount performance across customer segments, evaluated promotional strategies through A/B testing, and identified optimal discount offers that increased purchase conversion rate by 10% while improving average order value (AOV) by 5%. [Template example.]

My own resume bullet for this project (use this one, it is defensible): Designed and analysed a pre-registered checkout A/B test following the worked example in this document - baseline 5%, MDE 1 point, 50/50 allocation, sample size about 8,160 per group - and reported a significant lift from 5% to 6% (+1 point, +20% relative, z = 3.10, p = 0.002, 95% CI +0.37 to +1.63 points) with a staged-rollout recommendation and guardrail checks.

How to Explain It in a Job Interview - STAR Method

Situation & Task (the setup): I built an end-to-end A/B testing project analysing a redesigned conversion funnel. The goal was to determine if a new checkout design improved purchase conversions compared with the existing baseline.

Action (the analytical rigour): I evaluated a sample of 20,000 users split evenly between Control (A) and Treatment (B). The baseline conversion rate was 5%, and the treatment achieved 6%. To ensure this was not just sampling variation, I calculated the standard error and constructed a 95% confidence interval for the absolute lift: +1.00 point, 95% CI +0.37 to +1.63 points (a +20% relative lift). As an additional step, I also ran a chi-square test to check whether user distributions across devices (mobile, tablet, and desktop) shifted significantly between variants.

Result & Business Takeaway (the decision): Because the p-value is less than 0.05 and the entire confidence interval is above zero, I confirmed the result was statistically significant and recommended rolling out Version B in a staged rollout, with continued monitoring, projecting an overall increase in purchase volume.

Interview tips: Do not fake real traffic. Say: 'This was a project I built to simulate an enterprise-scale experimentation pipeline.' Be prepared for follow-up questions: Why two-sided, not one-sided? What would you do if SRM appeared? What if the primary metric wins but AOV falls? Interviewers care far more about how cleanly you think about experimentation than whether the website had real users.

Follow-Up Questions to Rehearse

- Why did you choose a 1-point MDE, and what happens to sample size if you halve it? (It roughly quadruples.)
- Your test is significant at day 3 - do you stop? (No: fixed horizon, unless a pre-registered sequential rule was set.)
- Conversion is up, but refund rate is up too - do you ship? (No, or ship with a fix: guardrails decide, not the primary metric alone.)
- How do you handle users who see both versions? (Exclude them, or better, assign by visitor ID and keep assignment consistent.)

End. Sources: teaching example structure from 'AB Testing and Statistical Analysis by Rishabh Mishra' (Google Drive, shared file); calculations verified in Python/SciPy for this document; experiment is simulated.